

# NOVA PROGRAM

## Nova Cybersecurity Requirements

Domain: Cybersecurity

Revision A | Synthetic extraction benchmark

# Nova Cybersecurity Requirements

Legacy controlled requirement sheet

## **REQ-NVA-CYS-001 Secure boot**

At every startup, the controller shall verify the application signature before executing application code.

## **REQ-NVA-CYS-002 Authentication lockout**

After five consecutive invalid authentication attempts, diagnostic security access shall be locked for 15 minutes.

## **REQ-NVA-CYS-003 Update freshness**

The bootloader shall reject a correctly signed software image when its security version is lower than the installed version.

3. Verification-oriented requirements

| Identifier      | Requirement title  | Normative requirement                                                                                                                                                                    |
|-----------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQ-NVA-CYS-004 | SecOC verification | For protected CAN messages, the receiver shall verify freshness and message authentication before releasing the payload, and shall increment the intrusion counter after a failed check. |
| REQ-NVA-CYS-005 | Key handling       | Production private keys shall be generated inside the hardware security module, shall never be exported in plaintext, and shall be erased when ownership transfer is authorized.         |

Applicability notes

| Context                               | Interpretation                                                              |
|---------------------------------------|-----------------------------------------------------------------------------|
| Unless a requirement states otherwise | Limits apply to the complete production-intent assembly.                    |
| Use of shall                          | The complete sentence is normative; table columns do not split obligations. |

## 4. Normative installation figure

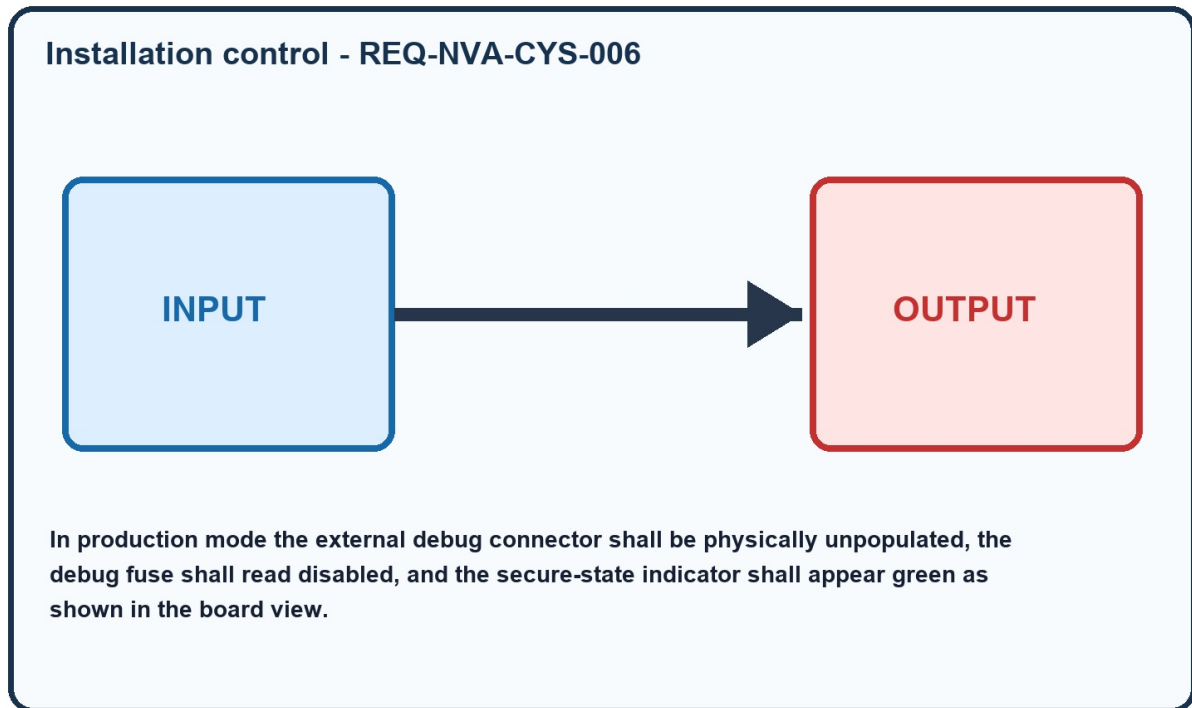


Figure 4-1. Colored geometry and embedded annotation are normative.

### Engineering note

The figure annotation forms part of the controlled requirement baseline.

Color, direction, sequence, and geometric relationships must be interpreted visually.